Literature Review for Selection of Base Model

Heart disease remains one of the leading causes of global mortality, demanding accurate and early prediction systems. With the advancement of Machine Learning (ML) and Deep Learning (DL), predictive models have significantly improved diagnostic accuracy and decision support systems in cardiology.

Recent studies show that traditional classifiers such as Support Vector Machine (SVM) and Random Forest (RF) remain widely used due to their robustness and interpretability [8]. However, more advanced approaches such as Stacked Ensemble Learning [1], Blending-based Networks [5], and Deep Learning with Transfer Learning [3] have demonstrated superior performance in certain datasets.

Binary classification models generally achieve higher accuracy compared to multiclass prediction models, which remain more complex and challenging [1], [6]. Furthermore, preprocessing techniques such as **Principal Component Analysis (PCA)** and optimized feature selection have been shown to significantly improve model performance [2].

This review analyzes and compares major machine learning techniques used for heart disease prediction to identify suitable base models for further experimentation.

Research Gaps Indetified

Based on the reviewed literature, the following research gaps are observed:

1. **Lower Performance in Multiclass Classification** – Multiclass models show significantly lower accuracy compared to binary classification models [1], [6].

2. **Dependence on Dataset Quality** – Model performance varies across datasets such as Cleveland, UCI, and regional datasets, indicating limited generalizability [4], [8].

3. **Limited Benchmarking Across Models** – Many studies compare limited algorithms rather than performing extensive benchmarking [8].

4. **Need for Dimensionality Reduction** – High-dimensional medical datasets require preprocessing techniques like PCA for better performance [2].

5. **Ensemble Complexity** – Although stacking and blending improve accuracy, they increase model complexity and computational cost [1], [5].

6. **Limited Explainability and Real-World Deployment** – Few models integrate explainable AI or deploy practical web-based systems [9].

These gaps suggest that starting with strong baseline models before moving to complex ensemble architectures is a systematic and practical approach.

Comparison Table of the Models used in the Papers

Ref No.	Study	Models Used	Best Performing Model	Accuracy	Key Techniques Used
[1]	Stacked Ensemble (Binary vs Multiclass)	SVM [1], RF [1], XGBoost [1], Gradient Boosting [1], Deep Learning [1], Stacking Ensemble [1]	Stacking Ensemble [1]	90.2% (Binary), 64.1% (Multiclass)	PCA, Stacked Ensemble
[2]	Comparative Multiclass Analysis	SVM [2], RF [2], Logistic Regression [2], XGBoost [2]	XGBoost [2]	Improved with preprocessing	Feature Selection, PCA
[3]	DL & Transfer Learning Review	CNN [3], RNN [3], Hybrid DL Models [3]	Deep Learning Models [3]	High accuracy (varies)	Transfer Learning
[4]	Indonesia Multiclass Study	SVM [4], RF [4], Decision Tree [4]	Dataset-dependent [4]	Moderate	Multiclass Classification
[5]	EnsCVDD-Net & BICVDD-Net	Ensemble Networks [5], Blending Models [5]	Blended Ensemble [5]	Improved over traditional ML	Blending & Ensemble
[6]	Stacked Ensemble Comparison	RF [6], SVM [6], LGBM [6], GBC [6], Stacking [6]	Stacking [6]	Higher than individual models	Ensemble Learning
[7]	ML Classification Techniques	SVM [7], KNN [7], DT [7], RF [7]	Hybrid Model [7]	Improved performance	Feature Engineering
[8]	Systematic Literature Review	SVM [8], RF [8], KNN [8], NB [8]	No single best	Dataset dependent	Preprocessing & Ensemble
[9]	Streamlit-Based System	RF [9], XGBoost [9], DT [9], KNN [9]	Random Forest [9]	95% (Heart Disease)	Web Deployment

Conclusion: From the literature analysis:

- SVM is consistently used across multiple studies and performs reliably across datasets [1], [2], [4], [8].
- MLP (Multi-Layer Perceptron), as part of deep learning approaches, shows strong potential in learning complex nonlinear relationships [3].
- Random Forest is robust, interpretable, and performs well in multiple comparative studies [1], [8], [9].

Although ensemble and stacking models achieve higher accuracy [1], [5], [6], it is methodologically sound to first establish strong baseline models before implementing complex architectures.

Therefore, for baseline models, we propose:

1. Support Vector Machine (SVM)
2. Multi-Layer Perceptron (MLP)
3. Random Forest (RF)

References Used:

[1] Improving Heart Disease Prediction with Stacked Ensemble Learning: A Comparison of Binary and Multiclass.

[2] Comparative Analysis of Multiclass Heart Disease Prediction Classification Models using Preprocessing and Feature Selection.

[3] Deep Learning and Transfer Learning in Cardiology: A Review of Cardiovascular Disease Prediction Models.

[4] Heart Disease Prediction in Indonesia Using Machine Learning Techniques for Multiclass Classification.

[5] Heart Disease Prediction Using Novel Ensemble and Blending Based Cardiovascular Disease Detection Networks (EnsCVDD-Net and BlCVDD-Net).

[6] Improving Heart Disease Prediction with Stacked Ensemble Learning: A Comparison of Binary and Multiclass.

[7] Predicting Heart Diseases Using Machine Learning and Different Data Classification Techniques.

[8] Survey of the Performance of Classification Algorithms on Heart Disease Datasets: A Systematic Literature Review.

[9] Streamlit-Based Multi-Disease Prediction System for Early Detection of Kidney Disease, Breast Cancer, and Heart Disease.